

Monalisa Portfolio

Design System & UI/UX Documentation

A record of the visual design language, layout architecture, component system, animation strategy, and responsive rules used to build the portfolio at [D:\PortlMonalisa-Portfolio](#). This documents the **design decisions**, not the personal/resume content (that lives separately in `src/data/portfolioData.js`).

1 Design Concept

Theme: "Deep-space AI/ML" identity — a near-black canvas with an electric **violet** → **cyan** accent gradient, chosen to read as GenAI/Data-Science rather than generic "developer portfolio" red/blue. The palette was picked specifically to complement the hero portrait video, which already contains violet neural-network holograms and cyan/blue code overlays in its background — so the UI feels like it was designed around the footage, not bolted onto it.

Reference architecture: built from an existing single-page portfolio template (Preloader → Navbar → Hero → content sections → Contact → Footer), restyled end-to-end with a new palette, new typography pairing, and content sections remapped to a Data Scientist / GenAI narrative instead of a full-stack-developer narrative.

2 Color System

All colors are defined once and reused via Tailwind utility classes and two CSS helper classes (`.text-gradient`, `.font-display` / `.font-code`) in `src/index.css`.

	Base background #08070d	Primary page background — near-black, used on Hero, Skills, GenAI, Experience, Soft Skills, Footer sections.
	Secondary background #0a0912	Slightly lighter near-black used to alternate section rhythm (Approach, Projects, Internships) so sections read as distinct panels without hard borders.
	Card / surface fill #0d0b16	Fill color inside gradient-bordered cards (About photo frame, hero video card, floating code chips, Preloader background elements).
	Violet — primary accent #7c3aed	Primary brand accent. Used in gradient text, button gradients, glows, borders, active states, timeline dots.
	Violet light #a78bfa	Lighter violet used as the gradient's starting stop (text-gradient, card borders) for a softer top-edge highlight.
	Cyan — secondary accent #22d3ee	Secondary accent, always paired with violet in gradients (never used alone) — represents the 'data/logic' half of the AI identity.



Reserved exclusively for live-status signals: the pulsing 'Open to opportunities' dot and success states (form submitted, footer availability line).

Gradient formula (the single reusable "brand gradient", 100° angle): `linear-gradient(100deg, #a78bfa 0%, #7c3aed 45%, #22d3ee 100%)` — applied to headline spans via the `.text-gradient` class, and to buttons/borders as `bg-gradient-to-r from-violet-600 to-cyan-500`.

Text opacity ladder (dark sections): white text is never pure white for body copy — a consistent opacity scale creates hierarchy without introducing new colors: `text-white` (headings) → `text-white/70-80` (body) → `text-white/40-60` (meta/labels/timestamps).

3 Typography

Three Google Fonts are loaded and assigned distinct jobs so the type system itself communicates structure:

- **Inter** (300–900) — the default body font (set on `<body>`). Used for paragraphs, descriptions, and any long-form reading text.
- **Space Grotesk** (400–700) — assigned via `.font-display`. Used for every heading, section title, the logo wordmark, and card titles — gives the site a distinct "engineered" display voice instead of relying on Inter everywhere.
- **JetBrains Mono** (400–700) — assigned via `.font-code`. Used for anything that should feel like "system output": badges/pills (e.g. "MY APPROACH"), skill percentages, timestamps, tech-tag chips, and the floating code-snippet chips.

Scale discipline: section headings are consistently `text-4xl md:text-5xl font-black` (or `lg:text-6xl` for hero/feature headings), badges/eyebrows are always a pill: `text-xs font-bold uppercase tracking-widest` inside a bordered, blurred pill. This repetition is what makes 13 different sections feel like one system.

4 Layout Architecture

Single-page app, 15 stacked sections rendered in `App.jsx` in this order:

- Preloader (overlay, not a section) — Navbar (fixed) — Hero — About — Impact Stats band — Technical Skills — My Approach — GenAI Expertise — Projects — Work Experience — Internships — Education & Certifications — Soft Skills — Contact — Footer.

Section rhythm: every full-bleed section alternates between the two dark background tones (`#08070d` / `#0a0912`) and, where content needs to "pop", switches to a bright violet block (Education & Certifications, Contact form) — the same technique the original reference template used with red blocks, re-themed to violet.

Consistent section anatomy used across almost every content section: an ambient blurred glow (violet or cyan, `blur-[120-160px]`, low opacity) positioned absolutely behind the content; a small bordered/blurred "eyebrow" pill; a large `font-display font-black` heading with one phrase wrapped in `.text-gradient`; a supporting paragraph capped at a max-width; then the section's content grid.

5 Hero Section — Design Rationale

The hero went through the most design iteration because the source video is a **vertical/portrait phone recording** (1512×2688, 9:16), which does not natively fit a widescreen hero background.

Problem with the first approach

Initial version forced the portrait video to fill the entire widescreen hero using `object-cover`, which cropped hard into just the subject's mouth/chin on desktop and looked dark/underexposed.

Final approach — build the layout around the video, not the reverse

- **Media-card pattern:** instead of a full-bleed background video, the portrait video lives inside a dedicated, self-contained card — identical in construction to the About-section photo card (2px gradient border, rounded-3xl, dark fill, drop shadow) — so the hero reuses a component pattern instead of inventing a new one.
- **Aspect ratio is mathematically preserved:** the card is set with `CSS aspect - [9/16]` so the video is never stretched, squashed, or cropped out of proportion; `object-fit: cover` with `object-position: center` keeps the subject centered within that correctly-proportioned frame.
- **One fluid sizing formula, not per-device breakpoints:** `width: min(90vw, 460px, calc((100vh - 240px) * 9 / 16))` — the card is capped by viewport width, a maximum premium width (460px), and available viewport height simultaneously. This guarantees the hero never forces a page scroll on short laptop screens while staying large and premium on 1920px displays.
- **Layout order flips responsively:** a flex container stacks the media card *above* the text on mobile/tablet (`order -1`) and moves it to the *right* of the text on desktop (`lg: order -2`) — single JSX tree, no separate mobile/desktop markup.
- **Exposure tuned to blend, not dominate:** the video inside the card is very lightly adjusted (`brightness(0.96)` `contrast(1.03)` `saturate(1.06)`) so it sits naturally in the dark theme rather than looking like a pasted-in photo.
- **In-card controls:** mute/unmute and play/pause are small glassmorphic circular buttons pinned to the bottom-right corner of the card itself (not floating over the whole hero), so they read as part of the "media player" object.
- **Floating code chip:** a small glassmorphic chip ("`import torch / > model ready`") overlaps the bottom-left corner of the card with a slow float animation — the same decorative device used on the About photo card ("`model.fit(X, y) / accuracy: 0.94`"), reinforcing that both cards belong to one system.

Text column

Widened to `max-w-2xl` so the two-line headline wraps naturally at large sizes (`text-5xl` on mobile up to `text-7xl` on xl screens) without orphaned words; the layout switched from CSS Grid to a **flex cluster** specifically to remove dead empty space between the text and the media card that a fixed-fraction grid column had left on mid-size laptop screens.

6 Component System

15 React components, one responsibility each, all reading shared copy from a single data file (`src/data/portfolioData.js`) so content and presentation are fully decoupled:

- **Preloader** — full-screen violet overlay; wordmark fills in via a Framer Motion clip-path reveal, then the whole panel slides up and out.
- **Navbar** — fixed, transparent at top, gains a blurred dark background past 50px scroll; underline-on-hover nav links; mobile slide-down menu.
- **Hero** — see Section 5.
- **About** — gradient-bordered photo card with a floating code chip; three tech "tiles" (Python / PyTorch-TF / LLMs) with icon + label.

- **Stats** — four-column impact-metric band with a scroll-triggered count-up animation (IntersectionObserver + requestAnimationFrame easing).
- **TechnicalSkills** — six category cards, each listing skills as animated gradient progress bars.
- **Approach** — four-step process cards connected by an SVG dashed path whose reveal is driven by scroll progress (Framer Motion useScroll + pathLength); each card highlights (fills violet) once the animated line reaches it.
- **GenAIExpertise** — 2×2 grid of icon cards (RAG, Agentic AI, Fine-tuning, NLP/Document AI).
- **Projects** — stacked project cards; the flagship project gets a stronger gradient border treatment to visually outrank the others.
- **Experience** — central vertical timeline with alternating left/right cards and a glowing gradient dot per entry.
- **Internships** — two-column card grid with skill/tech tag lists and outbound repo links.
- **EducationCerts** — the "bright" violet-to-cyan gradient block (mirrors the reference template's red accent block) pairing one education card with a certificate grid.
- **SoftSkills** — 4-column icon-card grid, dark background, consistent with TechnicalSkills card styling.
- **Contact** — huge outlined background wordmark with scroll parallax (Framer Motion useTransform), overlaid by a solid violet form panel; EmailJS submission with a prefilled-mailto fallback if EmailJS isn't configured.
- **Footer** — three-row layout (taglines / credential / availability, huge lowercase wordmark, contact + socials) matching the reference template's footer structure exactly, re-themed.
- **SocialCons.jsx** — a shared file of inline SVG icon components (GitHub, LinkedIn, Mail, Arrow, ExternalLink) so every section reuses the exact same icon markup instead of duplicating SVG paths.

7 Motion & Interaction

- **AOS (Animate-On-Scroll)**: the default reveal system across the whole site — fade-up / fade-left / fade-right / zoom-in on nearly every section header, card, and list item, each with staggered data-aos-delay so multi-item grids cascade in rather than popping in together.
- **Framer Motion**: reserved for animation that AOS can't express — the Preloader's clip-path text fill and slide-out exit, the Approach section's scroll-synced SVG line draw, and the Contact section's background-text scroll parallax.
- **Custom keyframes** (defined once in index.css / Tailwind theme): float (6s bob, used on floating code chips), gridPan (20s panning dot-grid background), and dropBounce (a bespoke overshoot-bounce entrance reserved for the About section's ID-badge photo).
- **Micro-interactions**: buttons scale 1.05–1.1 and gain a colored glow shadow on hover; cards lift with hover:scale-[1.02–1.03] plus a soft violet drop-shadow; nav links get an animated underline; the "available" status dot uses a CSS animate-ping halo.

8 Responsive Strategy

One JSX tree per component — no separate mobile/desktop component variants. Responsiveness is handled entirely through:

- Tailwind breakpoint prefixes (sm/md/lg/xl/2xl) that change font-size, spacing, grid column count, and element order.
- CSS min() / calc() formulas for elements that need true fluid sizing (the hero media card) instead of a fixed set of breakpoint widths.
- Flex order utilities to reflow the same markup (e.g., hero image above text on mobile, beside text on desktop).

- A mobile-specific hamburger menu in the Navbar with its own slide-down panel, sharing the same nav-link data as desktop.

Verified with real DOM measurements (not visual guesswork) at four breakpoints: 1920×1080 desktop, 1366×768 laptop, 768×1024 tablet, and 375×812 mobile — checking aspect-ratio integrity, absence of horizontal overflow, and that the hero never forces vertical page-scroll on short screens.

9 Tech Stack

- **React 19 + Vite 8** — component framework and dev/build tooling.
- **Tailwind CSS v4** (via [@tailwindcss/vite](https://tailwindcss.com/docs/using-v4)) — all styling, no separate CSS-in-JS.
- **Framer Motion** — the Preloader, the Approach section's scroll-driven line, and the Contact parallax.
- **AOS** — the site-wide scroll-reveal library.
- **EmailJS** — optional contact-form email delivery (gracefully falls back to a prefilled `mailto:` if unset).

10 Where Things Live

File	Role
<code>src/index.css</code>	Color tokens (base bg), font-family assignments, gradient/scrollbar/selection helpers, custom keyframes.
<code>src/data/portfolioData.js</code>	All personal/resume content — kept separate from every component so text edits never touch design code.
<code>src/components/*.jsx</code>	One file per section (see Section 6).
<code>src/components/SocialIcons.jsx</code>	Shared inline-SVG icon set reused across Hero, Navbar, Footer, Projects, Contact.
<code>tailwind / vite config</code>	Standard Vite + Tailwind v4 plugin setup, no custom Tailwind theme overrides beyond the animation keyframes in <code>index.css</code> .

Generated as a design-system reference for the Monalisa Burma portfolio project.